

(1) Warehousing Robot (DP & value Iteration)

Dynamic programming is a technique that approaches bellman equation & by using value iteration it improves the efficiency. The Robot needs to travel from source to destination Working

1. Each States : Goal, warehouse, obstacles are assigned
2. Action : based on actions like up, down, left, right it will go.
3. By using value iteration (getting reward assigned),
Uses:-

1. Find the Shortest path
2. Reduces the cost. Therefore, by using DP & value Iteration we can build a cost-efficient Agent.

(2) Recommendation System (Monte Carlo Method)

By using policy optimization (Monte Carlo method) -
- is used to recommend the customers (ex. Netflix).

Working

1. First it observe the Customer State.
2. It will recommend and accordingly get the reward.
3. By using the policy it will maximize the rewards (optimize)

Uses:-

1. For long-run Customer Recommendation.
2. Policy optimization.

By using this monte-carlo method we can optimize the policy according to Customer actions.

(3) Delivery - Robot (SARSA ALGORITHM)

The Robot has to deliver in uncertain environment by using SARSA.

State $\rightarrow$ Action $\rightarrow$ Reward $\rightarrow$ State $\rightarrow$ Action.

Working :-

1. It will observe the state and based on that it will take the action, then get the reward.
2. By using Q-values it will update and then it will choose the best action.

Uses :-

1. Works for uncertain environment (Stochastic).
2. It will optimize the path.

(4) Robotic Arm (REINFORCE ALGORITHM)

Reinforce Algorithm is a policy gradient algorithm which uses continuous learning from the environment.

Working :-

1. It defines state and action, get reward by using probability.
2. It will continuously learn from the previous action and choose the best for the next step.
3. By using we can train the arm how to perform continuously.

Uses :-

1. Continuous learning.

Autonomous vehicle (Q-learning, DQN)

Q-learning $\rightarrow$ By using Q-values it will update.
DQN $\rightarrow$ It uses neural network and it interacts with the environment.

Working :-

1. Q-learning - It updates values by using Q-table like it can be used through large network.
2. DQN - It uses neural network and it will improve the rewards by interacting & observing the environment.

Uses :-

1. Q-learning - used for large network (Not complex)
2. DQN - used for complex training.

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Game AI (DQN, DDQN, Dueling DQN)

DQN $\rightarrow$ uses neural network to interact with environment.
DDQN $\rightarrow$ It separates the States $\rightarrow$ Selecting the state
 $\rightarrow$ Evaluating the state.
Dueling DQN $\rightarrow$ It is trained to handle complex tasks

Working

1. ^{DQN} By using NN and interaction with environment it will maximize (or) optimize the reward.
 2. DDQN - 1. By using (selecting the state) it will select.
2. It will evaluate the selected step based on this it will update the Q-values.
 3. Dueling DQN - By separating the environments it will be used for complex gaming.
- Uses :- 1. Computational efficiency.
2. policy optimization.

(7) Smart Drone Navigation (A2C, A3C)

A2C - Synchronous learning (works like a team)

A3C - Asynchronous learning (works independently and update accordingly).

Working :- (By using grid System)

1. A2C $\rightarrow$ A: Actor (It chooses ^(selects) the states by using grid)
C: critic (By working Synchronously it will reward)

2. A3C $\rightarrow$ A: Actor (Same)

C: critic (By working independently the data will be collected and reward is updated).

uses :-

1. optimal path.
2. computational efficiency.

(8) Healthcare System (POMDP)

POMDP : Partially Observed Markov decision process
It will identify incomplete tasks like healthcare.

Working :-

1. The patient results depends on various parameters like tests, scanning etc.
2. By using POMDP we can observe the each state and according to that it will reward by selecting best action performed.

uses :-

1. For incomplete tasks.
2. We can predict by using previous state and perform best action.

Robot Navigation (PPO, TRPO, DDPG)

PPO (Proximal Policy Optimization) - By using policy it will maximize rewards

TRPO (Trust Region Policy Optimization) - By observing the environment it will restrict under the trust region.

DDPG (Deep Deterministic Policy Gradient)

By using the policy gradient it will select the action and get the reward.

Working:-

By using this we can optimize the policy for Robot navigation.

1. PPO - By using 1-step process - it will select the best action by using the policy.
2. TRPO - The Robot will navigate by interacting with environment it will restrict it in Trust Region to maximize the reward (like grid).
3. Deep-Deterministic Policy Gradient (DDPG)
By determining the policy it will maximize the reward.

Uses:-

1. Policy optimization.
2. Computation efficiency.

400 MARL (Multi-Agent Reinforcement Learning in Traffic Management)

In this MARL Algorithm Agent-A₁ plays an important role by interacting with multiple agents.

Working:-

Agent : Traffic A₁

State : Red, green, orange

Action : By using environment.

1. By interacting with environment and multiple agents it will decide the best action.
2. To find the best action to reduce the problems.

Challenges:-

1. Interacting with environment.
2. Multiple agents.

Uses:-

1. Reduces vehicle waiting time.
2. Traffic congestion.

The MARL is very useful in cities to reduce the problems.